

MicroOS-Sim – Test Week 3

Week 3 tests for Memory, Stack, FIFO Queue and the new simulator operations.

Test ID	Test	Expected Result	Actual Result	Status
TC01	Memory Write / Read	Write ABH at 0AH and read ABH.	Read returned ABH at 0AH.	PASS
TC02	Memory Reset	Written value becomes 00H after reset.	Address 0AH returned 00H.	PASS
TC03	PUSH / POP LIFO	Push 10H,20H; POP returns 20H then 10H.	Returned 20H then 10H.	PASS
TC04	Stack Pointer	SP returns to 07H after POP.	SP = 07H.	PASS
TC05	Enqueue / Full	10,20,30 fills a size-3 queue.	Queue = 3/3 FULL.	PASS
TC06	Dequeue / FIFO	Remove 10, then 20, then 30.	Returned 10,20,30.	PASS
TC07	Empty condition	Queue is empty after all three dequeues.	Queue reported EMPTY.	PASS
TC08	Simulator Queue	ENQUEUE 10,20; DEQUEUE → R0=10,R1=20.	R0=10, R1=20.	PASS
TC09	Simulator PUSH / POP	PUSH 10,20; first POP returns 20.	POP returned 20.	PASS

Queue Assembly Validation

Program: ENQUEUE 10, ENQUEUE 20, ENQUEUE 30, DEQUEUE into R0, R1 and R2.

Expected Result	Actual Result	Status
R0=10; R1=20; R2=30; FIFO order preserved.	R0=10; R1=20; R2=30; FIFO order preserved.	PASS

Overall Testing Result

9/9 Week-3 functional cases PASS. Queue Assembly validation PASS. Memory Read/Write and reset, Stack PUSH/POP and SP, FIFO ordering, empty/full conditions and simulator-level new operations were verified.